

Normal Distribution:

A continuous RV X is said to have a normal distribution with mean μ and variance σ^2 ($N(\mu, \sigma^2)$) if it has pdf given by

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty, \\ \mu \in \mathbb{R}, \sigma > 0$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\text{Put } \frac{x-\mu}{\sigma} = t \\ \Rightarrow \frac{1}{\sigma} dx = dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt$$

$$= \frac{1}{\sqrt{2\pi}} 2 \int_0^{\infty} e^{-\frac{t^2}{2}} dt$$

$$\frac{t^2}{2} = z \\ \Rightarrow t dt = dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-z} \frac{1}{\sqrt{2z}} dz$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-z} z^{-\frac{1}{2}} dz$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-z} z^{\frac{1}{2}-1} dz$$

$$= \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = 1$$

$$X \sim N(\mu, \sigma^2)$$

$$M_X(t) = E(e^{tX})$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$dz = \frac{1}{\sigma} dx$$

$$z = \frac{x-\mu}{\sigma}$$

$$= \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{t\sigma z - \frac{z^2}{2}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2t\sigma z)} dz$$

$+ t^2\sigma^2 - t^2\sigma^2$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\{(z - t\sigma)^2 - t^2\sigma^2\}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - t\sigma)^2} \cdot e^{\frac{1}{2}\sigma^2 t^2} dz$$

$$= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - t\sigma)^2} dz$$

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$M'_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot (\mu + \sigma^2 t)$$

$$M''_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot (\mu + \sigma^2 t)^2 + e^{\mu t + \frac{1}{2}\sigma^2 t^2} \sigma^2$$

$$M'_X(0) = \mu = E(X)$$

$$M''_X(0) = \mu^2 + \sigma^2 = E(X^2)$$

$$\begin{aligned}\therefore \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \mu^2 + \sigma^2 - \mu^2 \\ &= \sigma^2\end{aligned}$$

$$\therefore \text{standard deviation} = +\sqrt{\text{Var}(X)} = \sigma$$

$$\begin{aligned}E\left(\frac{X-\mu}{\sigma}\right)^k &= \int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right)^k \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \int_{-\infty}^{\infty} z^k \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = 0 \quad \text{if } k \text{ is odd,} \quad \frac{x-\mu}{\sigma} = z\end{aligned}$$

If k is even, say $k=2m$

$$\begin{aligned}&= 2 \int_0^{\infty} z^{2m} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \\ &= 2 \int_0^{\infty} (2t)^m \cdot \frac{1}{\sqrt{2\pi}} e^{-t} \frac{1}{\sqrt{2t}} dt \quad \begin{array}{l} \frac{z^2}{2} = t \\ \Rightarrow z dz = dt \end{array}\end{aligned}$$

$$= \frac{2^m}{\sqrt{\pi}} \int_0^{\infty} t^{m-\frac{1}{2}} e^{-t} dt$$

$$= \frac{2^m}{\sqrt{\pi}} \int_0^{\infty} t^{(m+\frac{1}{2})-1} e^{-t} dt = \frac{2^m}{\sqrt{\pi}} \Gamma(m+\frac{1}{2})$$

$$= \frac{2^m}{\sqrt{\pi}} \Gamma(m - \frac{1}{2} + 1)$$

$$= \frac{2^m}{\sqrt{\pi}} (m - \frac{1}{2}) \Gamma(m - \frac{1}{2})$$

$$= \frac{2^m}{\sqrt{\pi}} (m - \frac{1}{2}) (m - \frac{3}{2}) \dots \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}$$

$$E\left(\frac{x-\mu}{\sigma}\right)^{2m} = (2m-1)(2m-3)\dots 5 \cdot 3 \cdot 1$$

$$k=1 \text{ gives } E\left(\frac{x-\mu}{\sigma}\right) = 0 \Rightarrow \frac{1}{\sigma} E(x-\mu) = 0 \Rightarrow E(x) - \mu = 0 \\ \Rightarrow \underline{E(x) = \mu} \checkmark$$

$$k=2, \quad E\left(\frac{x-\mu}{\sigma}\right)^2 = 1 \Rightarrow E(x-\mu)^2 = \sigma^2 \\ \Rightarrow \text{Var}(x) = \sigma^2$$

$$k=3, \quad E\left(\frac{x-\mu}{\sigma}\right)^3 = 0$$

$$k=4, \quad E\left(\frac{x-\mu}{\sigma}\right)^4 = 3 \cdot 1 \Rightarrow \boxed{E(x-\mu)^4 = 3\sigma^4}$$

$$\mu_4 = 3\sigma^4$$

$$\beta_1 = 0, \\ \beta_2$$

$$\frac{\mu_4}{\mu_2^2} - 3 = \frac{3\sigma^4}{\sigma^4} - 3 = 0$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty, \quad \mu \in \mathbb{R}, \sigma > 0$$

$$\mu=0, \sigma=1$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty,$$

